

1. INTRODUCTION & PROBLEM STATEMENT

The objective of this project is to architect a robust relational database for a Food Delivery Aggregator ecosystem. The system is designed to seamlessly connect restaurants, customers, and delivery drivers. The database must maintain data integrity while tracking the entire lifecycle of an order from placement and food preparation to delivery status.

Database Challenge: The core complexity lies in managing multi-way relationships across Customer, Restaurant, Driver, and Order tables. The design also supports location information for drivers and operational hubs.

2. SYSTEM REQUIREMENTS

2.1 User Requirements vs. System Capabilities

- **User Requirements:** Customers need to browse menus and place orders securely. Restaurants require an interface to manage menu items and accept orders. Drivers need a system to receive dispatch requests and update their location.
- **System Capabilities:** The database uses Extended ER (EER) concepts, including specialization/generalization, to manage a central User superclass and Customer, Restaurant, and Driver subclasses. The system also supports multi-party transactions.

2.2 Functional Requirements (Data-Centric)

- The system must allow a Driver to update the status of an order, for example from PICKED_UP to DELIVERING.
- The system must support calculating order totals from OrderItem quantities and recorded item prices.
- Foreign-key relationships must preserve referential integrity between users, customers, restaurants, drivers, orders, menus, payments, and related entities.

2.3 Non-Functional Requirements

- **Performance:** Multi-way joins involving Orders, Users, and OrderItems should execute efficiently for dashboard and reporting workloads.
- **Security:** Sensitive customer data, particularly passwords and payment-related information, should be protected appropriately.
- **Integrity:** Relational mappings should follow normalization principles and minimize unnecessary redundancy.

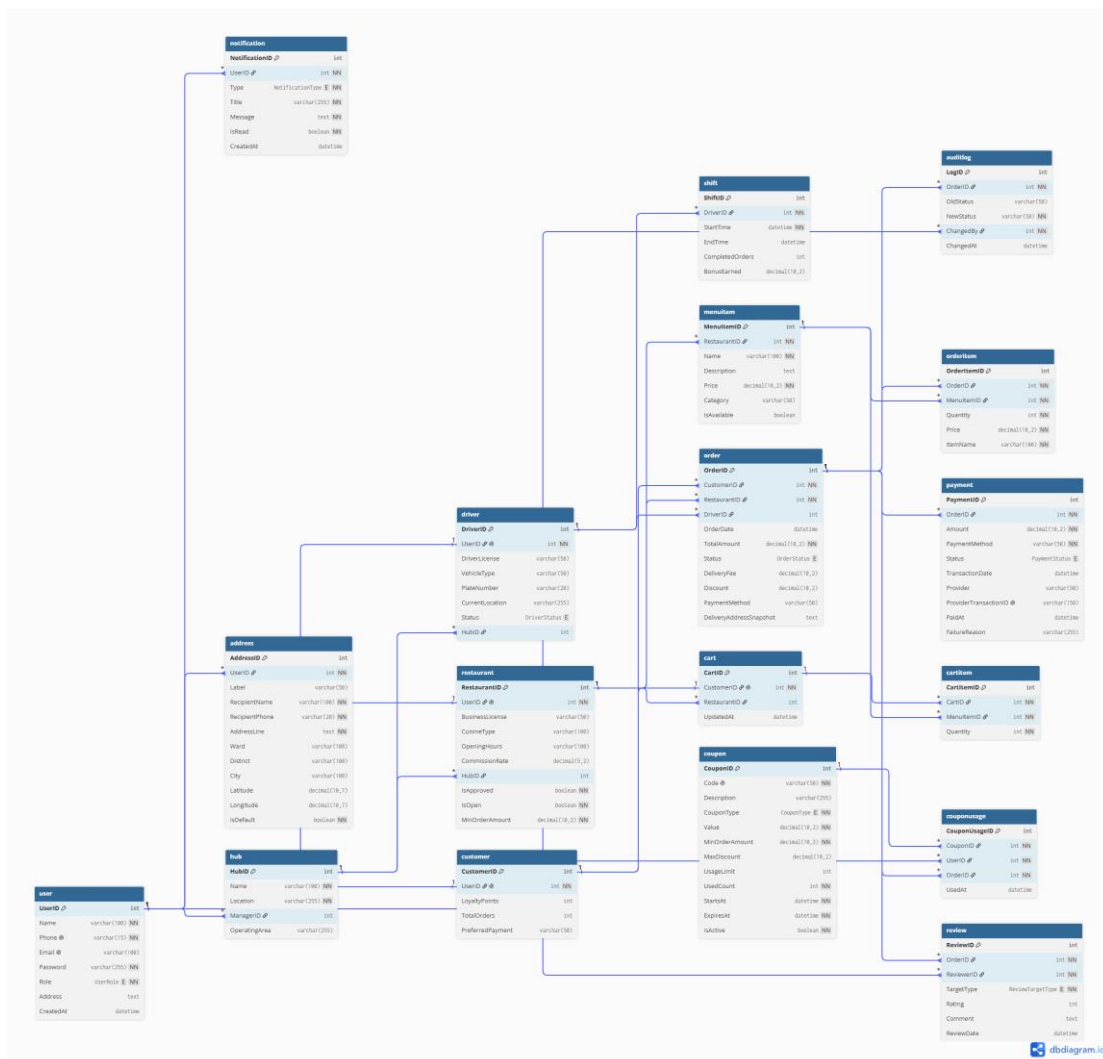
3. BUSINESS RULES

The following business rules describe the intended system behavior. Where a rule is not directly

enforced by the current SQL schema, it should be treated as a business requirement rather than a native database constraint.

- **BR-01 (Account Uniqueness):** A single phone number or email address can only be associated with one User account. This is enforced by UNIQUE constraints in the User table.
- **BR-02 (Operational Shifts):** A Driver should only be assigned to a new Order when their current status is ONLINE. The current SQL does not contain a CHECK constraint or trigger that enforces this condition.
- **BR-03 (Order Integrity):** An OrderItem must reference a valid MenuItem. The current SQL enforces the foreign key, but it does not enforce $Quantity > 0$ through a CHECK constraint.
- **BR-04 (Historical Retention):** Audit logs should be retained even if the user who initiated the change is removed. The current SQL uses a foreign key from AuditLog.ChangedBy to User.UserID without ON DELETE CASCADE or SET NULL; therefore, deletion behavior is not equivalent to automatic historical retention.
- **BR-05 (Commission Structure):** The intended default commission rate is 15%. However, the current SQL defines CommissionRate as nullable and does not specify a DEFAULT 15.00 constraint.

4. ENTITY-RELATIONSHIP (ER) DIAGRAM & DATA DICTIONARY



4.1 User Inheritance Structure (IS-A Hierarchy)

The system implements a Superclass/Subclass architecture following EER principles. User acts as the central superclass, while Customer, Restaurant, and Driver represent specialized user roles. Admin does not have a separate table and is represented by User.Role = 'Admin'.

4.1.1 User – Superclass

Definition: Central entity for identity, contact, authentication, and roles.

Main attributes: UserID, Name, Phone, Email, Password, Role, Address, CreatedAt.

Relationship: Referenced by Address, Hub.ManagerID, AuditLog, CouponUsage, Notification, and Review.

Constraint / note: Centralizes shared user information.

4.1.2 Customer – Subclass

Definition: Represents customers who place food orders.

Main attributes: CustomerID, UserID, LoyaltyPoints, TotalOrders, PreferredPayment.

Relationship: User 1:0..1 Customer; Customer 1:N Order; Customer 0..1 Cart.

Constraint / note: CustomerID and UserID are identifiers; UserID is UNIQUE.

4.1.3 Restaurant – Subclass

Definition: Represents restaurant partners providing food.

Main attributes: RestaurantID, UserID, BusinessLicense, CuisineType, OpeningHours, CommissionRate, HubID, IsApproved, IsOpen, MinOrderAmount.

Relationship: User 1:0..1 Restaurant; Restaurant 1:N MenuItem and Order; Restaurant N:1 Hub.

Constraint / note: UserID is UNIQUE; HubID is nullable.

4.1.4 Driver – Subclass

Definition: Represents delivery partners.

Main attributes: DriverID, UserID, DriverLicense, VehicleType, PlateNumber, CurrentLocation, Status, HubID.

Relationship: User 1:0..1 Driver; Driver 1:N Order and Shift; Driver N:1 Hub.

Constraint / note: UserID is UNIQUE; DriverID in Order is nullable.

4.1.5 Admin

Definition: Administrative role represented in the User table.

Main attributes: User.Role.

Relationship: Identified by User.Role = 'Admin'.

Constraint / note: No separate Admin table exists in the current SQL.

4.1.6 Address

Definition: Stores delivery/contact addresses belonging to users.

Main attributes: AddressID, UserID, Label, RecipientName, RecipientPhone, AddressLine, Ward, District, City, Latitude, Longitude, IsDefault.

Relationship: User 1:N Address.

Constraint / note: Supports multiple saved addresses and a default address.

4.1.7 Hub

Definition: Represents an operational center/area.

Main attributes: HubID, Name, Location, ManagerID, OperatingArea.

Relationship: Hub 1:N Restaurant and Hub 1:N Driver; ManagerID references User.

Constraint / note: ManagerID is nullable.

4.1.8 Order

Definition: Represents an order created by a customer.

Main attributes: OrderID, CustomerID, RestaurantID, DriverID, OrderDate, TotalAmount, Status, DeliveryFee, Discount, PaymentMethod, DeliveryAddressSnapshot.

Relationship: Customer 1:N Order; Restaurant 1:N Order; Driver 0..1:N Order assignment.

Constraint / note: DriverID may be NULL.

4.1.9 AuditLog

Definition: Stores order-status change history.

Main attributes: LogID, OrderID, OldStatus, NewStatus, ChangedBy, ChangedAt.

Relationship: Order 1:N AuditLog; User 1:N AuditLog.

Constraint / note: Provides traceability for status changes.

4.1.10 Cart

Definition: Represents a customer's shopping cart.

Main attributes: CartID, CustomerID, RestaurantID, UpdatedAt.

Relationship: Customer 0..1 Cart; Cart 0..1 Restaurant.

Constraint / note: CustomerID is UNIQUE.

4.1.11 MenuItem

Definition: Represents a food item offered by a restaurant.

Main attributes: MenuItemID, RestaurantID, Name, Description, Price, Category, IsAvailable.

Relationship: Restaurant 1:N MenuItem.

Constraint / note: Stores menu item information and availability.

4.1.12 CartItem

Definition: Represents a menu item added to a cart.

Main attributes: CartItemID, CartID, MenuItemID, Quantity.

Relationship: Cart 1:N CartItem; MenuItem 1:N

CartItem.

Constraint / note: UNIQUE(CartID, MenuItemID).

4.1.13 Coupon

Definition: Represents a discount code.

Main attributes: CouponID, Code, Description, CouponType, Value, MinOrderAmount, MaxDiscount, UsageLimit, UsedCount, StartsAt, ExpiresAt, IsActive.

Relationship: Coupon 1:N CouponUsage.

Constraint / note: Code is UNIQUE; supports PERCENT and FIXED types.

4.1.14 CouponUsage

Definition: Records a coupon being used.

Main attributes: CouponUsageID, CouponID, UserID, OrderID, UsedAt.

Relationship: Coupon 1:N CouponUsage; User 1:N CouponUsage; Order 1:N CouponUsage.

Constraint / note: UNIQUE(CouponID, OrderID).

4.1.15 Notification

Definition: Represents a notification sent to a user.

Main attributes: NotificationID, UserID, Type, Title, Message, IsRead, CreatedAt.

Relationship: User 1:N Notification.

Constraint / note: Type is ORDER, PROMOTION, or SYSTEM.

4.1.16 OrderItem

Definition: Represents a specific menu item in an order.

Main attributes: OrderItemID, OrderID, MenuItemID, Quantity, Price, ItemName.

Relationship: Order 1:N OrderItem; MenuItem 1:N OrderItem.

Constraint / note: Stores item price and name at order time.

4.1.17 Payment

Definition: Represents a payment transaction associated with an order.

Main attributes: PaymentID, OrderID, Amount, PaymentMethod, Status, TransactionDate, Provider, ProviderTransactionID, PaidAt, FailureReason.

Relationship: Order 1:N Payment in the current SQL because OrderID is not UNIQUE.

Constraint / note: Status is PENDING, SUCCESS, or FAILED. ProviderTransactionID is UNIQUE.

4.1.18 Review

Definition: Represents a user's review of a restaurant or driver.

Main attributes: ReviewID, OrderID, ReviewerID, TargetType, Rating, Comment, ReviewDate.

Relationship: User 1:N Review; Order 1:N Review.

Constraint / note: TargetType is Restaurant or Driver; ReviewerID references User.UserID.

4.1.19 Shift

Definition: Represents a driver's work shift.

Main attributes: ShiftID, DriverID, StartTime, EndTime, CompletedOrders, BonusEarned.

Relationship: Driver 1:N Shift.

Constraint / note: Supports driver workforce and shift management.

4.1.20 Entity Summary

Entity Group	Entity	Role
Superclass	User	Shared user information
Subclass	Customer	Customer placing orders
Subclass	Restaurant	Restaurant providing food
Subclass	Driver	Delivery driver
Role in User	Admin	System administration
User Support	Address	Delivery addresses
Management	Hub	Operational center/area
Transaction	Order	Customer order
Tracking	AuditLog	Order-status history
Shopping	Cart / CartItem	Shopping cart and contents
Restaurant Menu	MenuItem	Food item
Promotion	Coupon / CouponUsage	Discount code and usage history
Notification	Notification	System/user notifications
Transaction Detail	OrderItem	Order-line details
Payment	Payment	Payment transaction
Feedback	Review	Restaurant/driver review
Driver Management	Shift	Driver work shift

Main inheritance structure: User => Customer, User => Restaurant, User => Driver. Admin is identified through User.Role = 'Admin'.

4.2. DATA DICTIONARY (SQL-ALIGNED)

The following dictionary is aligned with the current MySQL SQL schema. It documents the actual data types, keys, nullability, defaults, foreign keys, and uniqueness constraints. Where the earlier report described a rule that is not present in the SQL, the SQL definition takes precedence here.

Table: User

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
UserID	INT	PK	No	AUTO_INCREMENT	Unique system-wide user identifier.
Name	VARCHAR(100)	-	No	-	User's full name.
Phone	VARCHAR(15)	-	No	-	UNIQUE; user phone number.
Email	VARCHAR(100)	-	Yes	NULL	UNIQUE; optional email address.
Password	VARCHAR(255)	-	No	-	Password value; application should store it securely hashed.
Role	ENUM('Customer','Restaurant','Driver','Admin')	-	No	-	User role.
Address	TEXT	-	Yes	NULL	Legacy/general contact address stored directly on User.
CreatedAt	DATETIME	-	Yes	CURRENT_TIMESTAMP	Account creation timestamp.

Table: Address

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
AddressID	INT	PK	No	AUTO_INCREMENT	Unique address identifier.
UserID	INT	FK	No	-	References User(UserID) ON DELETE CASCADE.
Label	VARCHAR(50)	-	Yes	NULL	Address label.
RecipientName	VARCHAR(100)	-	No	-	Recipient name.
RecipientPhone	VARCHAR(20)	-	No	-	Recipient phone number.
AddressLine	TEXT	-	No	-	Main street/address details.
Ward	VARCHAR(100)	-	Yes	NULL	Ward.
District	VARCHAR(100)	-	Yes	NULL	District.
City	VARCHAR(100)	-	Yes	NULL	City.
Latitude	DECIMAL(10,7)	-	Yes	NULL	Latitude coordinate.
Longitude	DECIMAL(10,7)	-	Yes	NULL	Longitude coordinate.
IsDefault	TINYINT(1)	-	No	-	Indicates whether the address is marked as default.

Table: Customer

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
CustomerID	INT	PK	No	AUTO_INCREMENT	Unique customer identifier.
UserID	INT	FK	No	-	UNIQUE; references User(UserID). No ON DELETE clause is defined in the current SQL.
LoyaltyPoints	INT	-	Yes	NULL	Customer loyalty points.
TotalOrders	INT	-	Yes	NULL	Stored total-order count.
PreferredPayment	VARCHAR(50)	-	Yes	NULL	Preferred payment method.

Table: Hub

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
HubID	INT	PK	No	AUTO_INCREMENT	Unique hub identifier.
Name	VARCHAR(100)	-	No	-	Hub name.
Location	VARCHAR(255)	-	No	-	Hub location.
ManagerID	INT	FK	Yes	NULL	References User(UserID).
OperatingArea	VARCHAR(255)	-	Yes	NULL	Area managed by the hub.

Table: Restaurant

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
RestaurantID	INT	PK	No	AUTO_INCREMENT	Unique restaurant identifier.
UserID	INT	FK	No	-	UNIQUE; references User(UserID). No ON DELETE clause is defined.
BusinessLicense	VARCHAR(50)	-	Yes	NULL	Business license identifier.
CuisineType	VARCHAR(100)	-	Yes	NULL	Primary cuisine classification.
OpeningHours	VARCHAR(100)	-	Yes	NULL	Restaurant operating hours.
CommissionRate	DECIMAL(5,2)	-	Yes	NULL	Restaurant commission rate. Current SQL has no DEFAULT 15.00.
HubID	INT	FK	Yes	NULL	References Hub(HubID); no ON DELETE SET NULL clause is defined.
IsApproved	TINYINT(1)	-	No	-	Restaurant approval flag.
IsOpen	TINYINT(1)	-	No	-	Restaurant open/closed flag.
MinOrderAmount	DECIMAL(10,2)	-	No	-	Minimum order amount.

Table: Driver

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
DriverID	INT	PK	No	AUTO_INCREMENT	Unique driver identifier.
UserID	INT	FK	No	-	UNIQUE; references User(UserID). No ON DELETE clause is defined.
DriverLicense	VARCHAR(50)	-	Yes	NULL	Driver license identifier.
VehicleType	VARCHAR(50)	-	Yes	NULL	Vehicle type.
PlateNumber	VARCHAR(20)	-	Yes	NULL	Vehicle registration plate.
CurrentLocation	VARCHAR(255)	-	Yes	NULL	Current location text.
Status	ENUM('ONLINE','OFFLINE','BUSY')	-	Yes	NULL	Driver operational status.
HubID	INT	FK	Yes	NULL	References Hub(HubID); no ON DELETE SET NULL clause is defined.

Table: Order

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
OrderID	INT	PK	No	AUTO_INCREMENT	Unique order identifier.
CustomerID	INT	FK	No	-	References Customer(CustomerID).
RestaurantID	INT	FK	No	-	References Restaurant(RestaurantID).
DriverID	INT	FK	Yes	NULL	References Driver(DriverID); may be unassigned.

OrderDate	DATETIME	-	Yes	CURRENT_TIMESTAMP	Order creation timestamp.
TotalAmount	DECIMAL(10,2)	-	No	-	Stored total order amount.
Status	ENUM('PENDING','CONFIRMED','PREPARING','READY','PICKED_UP','DELIVERING','COMPLETED','CANCELLED')	-	Yes	NULL	Order lifecycle status.
DeliveryFee	DECIMAL(10,2)	-	Yes	NULL	Delivery fee.
Discount	DECIMAL(10,2)	-	Yes	NULL	Applied discount.
PaymentMethod	VARCHAR(50)	-	Yes	NULL	Order-level payment method.
DeliveryAddressSnapshot	TEXT	-	Yes	NULL	Address snapshot captured for the order.

Table: AuditLog

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
LogID	INT	PK	No	AUTO_INCREMENT	Unique audit-log identifier.
OrderID	INT	FK	No	-	References Order(OrderID).
OldStatus	VARCHAR(50)	-	Yes	NULL	Previous order status.
NewStatus	VARCHAR(50)	-	No	-	New order status.
ChangedBy	INT	FK	No	-	References User(UserID).
ChangedAt	DATETIME	-	Yes	CURRENT_TIMESTAMP	Status-change timestamp.

Table: Cart

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
CartID	INT	PK	No	AUTO_INCREMENT	Unique cart identifier.
CustomerID	INT	FK	No	-	UNIQUE; references Customer(CustomerID) ON DELETE CASCADE.
RestaurantID	INT	FK	Yes	NULL	References Restaurant(RestaurantID).
UpdatedAt	DATETIME	-	Yes	CURRENT_TIMESTAMP	Last cart update timestamp.

Table: MenuItem

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
MenuItemID	INT	PK	No	AUTO_INCREMENT	Unique menu-item identifier.
RestaurantID	INT	FK	No	-	References Restaurant(RestaurantID).
Name	VARCHAR(100)	-	No	-	Menu item name.
Description	TEXT	-	Yes	NULL	Menu item description.
Price	DECIMAL(10,2)	-	No	-	Current item price.
Category	VARCHAR(50)	-	Yes	NULL	Menu category.
IsAvailable	TINYINT(1)	-	Yes	NULL	Availability flag.

Table: CartItem

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
CartItemID	INT	PK	No	AUTO_INCREMENT	Unique cart-item identifier.
CartID	INT	FK	No	-	References Cart(CartID) ON DELETE CASCADE.

MenuItemID	INT	FK	No	-	References MenuItem(MenuItemID) ON DELETE CASCADE.
Quantity	INT	-	No	-	Quantity of the menu item in the cart.
UNIQUE(CartID, MenuItemID)	-	UQ	-	-	Prevents duplicate rows for the same menu item in one cart.

Table: Coupon

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
CouponID	INT	PK	No	AUTO_INCREMENT	Unique coupon identifier.
Code	VARCHAR(50)	-	No	-	UNIQUE discount code.
Description	VARCHAR(255)	-	Yes	NULL	Coupon description.
CouponType	ENUM('PERCENT','FIXED')	-	No	-	Discount calculation type.
Value	DECIMAL(10,2)	-	No	-	Discount value.
MinOrderAmount	DECIMAL(10,2)	-	No	-	Minimum order amount required.
MaxDiscount	DECIMAL(10,2)	-	Yes	NULL	Maximum discount amount.
UsageLimit	INT	-	Yes	NULL	Maximum permitted uses.
UsedCount	INT	-	No	-	Number of uses recorded.
StartsAt	DATETIME	-	No	-	Coupon start time.
ExpiresAt	DATETIME	-	No	-	Coupon expiration time.
IsActive	TINYINT(1)	-	No	-	Coupon activation flag.

Table: CouponUsage

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
CouponUsageID	INT	PK	No	AUTO_INCREMENT	Unique coupon-usage identifier.
CouponID	INT	FK	No	-	References Coupon(CouponID) ON DELETE CASCADE.
UserID	INT	FK	No	-	References User(UserID) ON DELETE CASCADE.
OrderID	INT	FK	No	-	References Order(OrderID) ON DELETE CASCADE.
UsedAt	DATETIME	-	Yes	CURRENT_TIMESTAMP	Coupon-use timestamp.
UNIQUE(CouponID, OrderID)	-	UQ	-	-	Prevents the same coupon from being recorded twice for one order.

Table: Notification

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
NotificationID	INT	PK	No	AUTO_INCREMENT	Unique notification identifier.
UserID	INT	FK	No	-	References User(UserID) ON DELETE CASCADE.
Type	ENUM('ORDER','PROMOTION','SYSTEM')	-	No	-	Notification category.
Title	VARCHAR(255)	-	No	-	Notification title.
Message	TEXT	-	No	-	Notification content.
IsRead	TINYINT(1)	-	No	-	Read/unread flag.
CreatedAt	DATETIME	-	Yes	CURRENT_TIMESTAMP	Notification creation timestamp.

Table: OrderItem

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
OrderItemID	INT	PK	No	AUTO_INCREMENT	Unique order-item identifier.
OrderID	INT	FK	No	-	References Order(OrderID).
MenuItemID	INT	FK	No	-	References MenuItem(MenuItemID).
Quantity	INT	-	No	-	Quantity ordered.
Price	DECIMAL(10,2)	-	No	-	Price recorded for the item in this order.
ItemName	VARCHAR(100)	-	No	-	Item name recorded at order time.

Table: Payment

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
PaymentID	INT	PK	No	AUTO_INCREMENT	Unique payment identifier.
OrderID	INT	FK	No	-	References Order(OrderID). Not UNIQUE in current SQL.
Amount	DECIMAL(10,2)	-	No	-	Payment amount.
PaymentMethod	VARCHAR(50)	-	No	-	Payment method.
Status	ENUM('PENDING','SUCCESS','FAILED')	-	Yes	NULL	Payment status.
TransactionDate	DATETIME	-	Yes	CURRENT_TIMESTAMP	Transaction timestamp.
Provider	VARCHAR(50)	-	Yes	NULL	Payment provider.
ProviderTransactionID	VARCHAR(150)	-	Yes	NULL	UNIQUE provider transaction identifier.
PaidAt	DATETIME	-	Yes	NULL	Successful payment timestamp.
FailureReason	VARCHAR(255)	-	Yes	NULL	Reason for payment failure.

Table: Review

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
ReviewID	INT	PK	No	AUTO_INCREMENT	Unique review identifier.
OrderID	INT	FK	No	-	References Order(OrderID).
ReviewerID	INT	FK	No	-	References User(UserID).
TargetType	ENUM('Restaurant','Driver')	-	No	-	Review target type.
Rating	INT	-	Yes	NULL	Rating value.
Comment	TEXT	-	Yes	NULL	Review comment.
ReviewDate	DATETIME	-	Yes	CURRENT_TIMESTAMP	Review creation timestamp.
UNIQUE(OrderID, ReviewerID, TargetType)	-	UQ	-	-	Prevents duplicate reviews for the same order, reviewer, and target type.

Table: Shift

Attribute	Data Type	Key	Nullable?	Default	Business Rules / Constraints
ShiftID	INT	PK	No	AUTO_INCREMENT	Unique shift identifier.
DriverID	INT	FK	No	-	References Driver(DriverID).
StartTime	DATETIME	-	No	-	Shift start time.
EndTime	DATETIME	-	Yes	NULL	Shift end time.
CompletedOrders	INT	-	Yes	NULL	Number of completed orders in the shift.

BonusEarned	DECIMAL(10,2)	-	Yes	NULL	Bonus earned during the shift.
-------------	---------------	---	-----	------	--------------------------------

Relationship Summary

- User => Customer / Restaurant / Driver: each subclass has a UNIQUE, NOT NULL UserID foreign key, so each subclass row corresponds to exactly one User; a User may have zero or one row in each subclass table. The current SQL does not enforce that the three subclass tables are mutually exclusive.
- Customer => Order: 1:N. Restaurant => Order: 1:N. Driver => Order: 1:N, with DriverID nullable on Order.
- Customer => Cart: 0..1 per customer because Cart.CustomerID is UNIQUE.
- Restaurant => MenuItem: 1:N. Hub => Restaurant: 1:N. Hub => Driver: 1:N.
- Cart => CartItem: 1:N. MenuItem => CartItem: 1:N.
- Order => OrderItem: 1:N. MenuItem => OrderItem: 1:N.
- Order => Payment: potentially 1:N in the current SQL because Payment.OrderID is not UNIQUE. If the business requirement is exactly one payment per order, add UNIQUE(OrderID).
- Order => Review: 1:N. User => Review: 1:N.
- Order => AuditLog: 1:N. User => AuditLog: 1:N.
- Coupon => CouponUsage: 1:N. User => CouponUsage: 1:N. Order => CouponUsage: 1:N.
- User => Notification: 1:N. User => Address: 1:N. Driver => Shift: 1:N.

5. PHASE 1 NEXT STEPS

Moving forward, the team will transition the requirements and SQL-aligned entity definitions into a formalized Extended Entity-Relationship (EER) diagram. The modeling should accurately represent the User IS-A hierarchy and the 1:N / optional relationships defined by the current schema.